

Introduction

Introduction to the Dataset

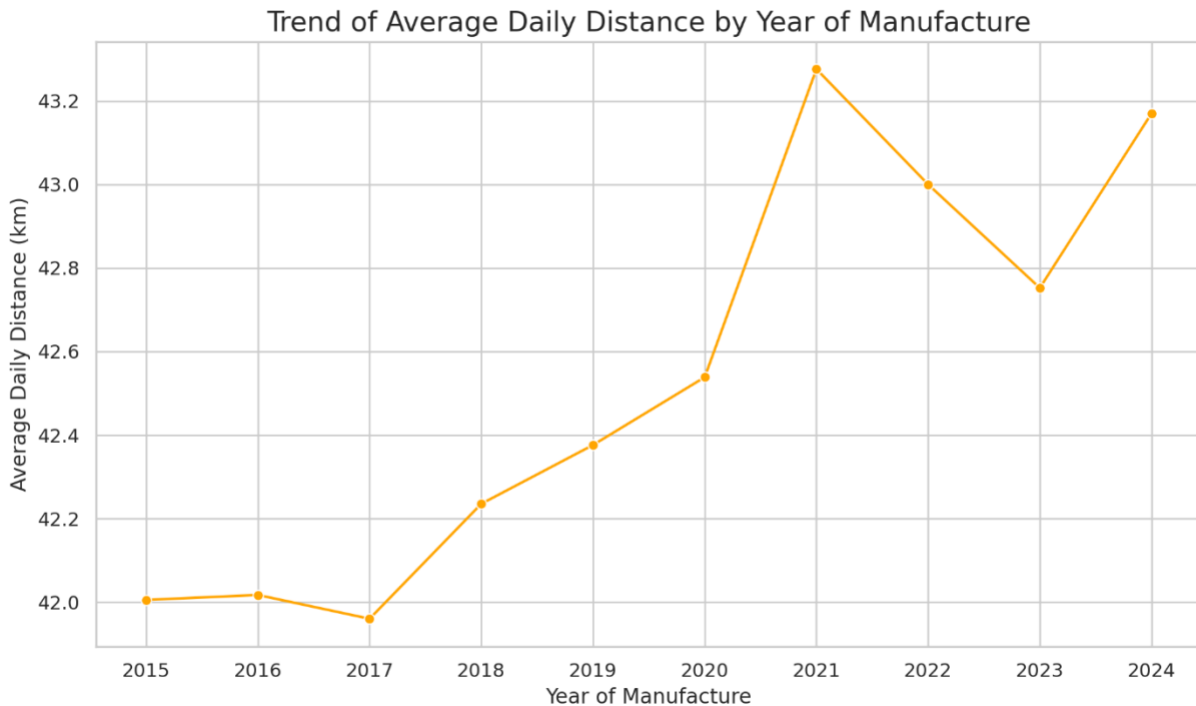
The dataset of bike sales in india sourced from KAGGLE contains information about bike sales in India, covering various attributes related to bike models, pricing, and ownership details. It includes **10,000 entries** with **15 columns**, providing insights into factors influencing the resale value and performance of different bike brands.

Key columns in the dataset include:

- **State & City Tier:** Location details where the bike was sold.
- **Brand & Model:** Manufacturer and specific bike model.
- **Price & Resale Price (INR):** Initial purchase price and estimated resale value.
- **Year of Manufacture & Registration Year:** Age and registration details of the bike.
- **Engine Capacity (cc) & Fuel Type:** Technical specifications affecting mileage and performance.
- **Mileage (km/l) & Avg Daily Distance (km):** Efficiency and usage statistics.
- **Owner Type & Seller Type:** Whether the bike is being sold by an individual or a dealer, and its previous ownership history.
- **Insurance Status:** Whether the bike is insured.

Data source: <https://www.kaggle.com/datasets/ak0212/indian-bike-sales-dataset>

Trend of Average Daily Distance by Year of Manufacture



This line graph shows the **trend of the average daily distance (in km) covered by motorcycles** based on their year of manufacture, ranging from **2015 to 2024**.

Key Observations:

From **2015 to 2017**, the average daily distance remained nearly constant, with **only minor fluctuations around 42 km**.

Starting in **2018**, there was a **gradual increase**, with the trend continuing upward until 2020.

The most significant jump **occurred in 2021**, reaching the highest recorded average.

After peaking in 2021, the **distance slightly declined in 2022 and 2023** before rising again in 2024.

Interpretation:

The increasing trend from **2018 to 2021** suggests that newer motorcycles are being used for **longer daily distances** compared to older models. This could be due to:

Better fuel efficiency & technology advancements in recent models.

Increased commuting distances due to urban expansion.

A rise in bike-based delivery services (e.g., food and e-commerce deliveries).

The **dip in 2022 and 2023** might be attributed to external factors such as:

Economic slowdowns or fuel price hikes.

A shift in commuting habits (e.g., work-from-home trends reducing daily travel).

The **recovery in 2024** suggests that motorcycles are being used more frequently again, possibly due to a **return to normal commuting patterns** or an increase in long-distance riders.

Potential Business Implications:

Bike manufacturers should focus on fuel efficiency and durability, given that newer bikes are being used for longer distances.

Insurance companies and maintenance service providers may need to adjust policies based on higher daily usage.

Rideshare and delivery businesses could consider expanding their fleet with **newer models** that can withstand higher daily usage.

This chart highlights the evolving **role of motorcycles in daily transportation**, with increasing distances signaling a growing dependence on two-wheelers for commuting and commercial purposes.

Bike Price Distribution by Brand



This report analyzes the distribution of bike prices across different brands using a box plot visualization. The box plot provides insights into price variations, central tendencies, and the overall price range of different brands in the Indian bike market.

- **Median (Central Line in the Box):** Represents the midpoint price, dividing the data into two equal halves.
- **Interquartile Range (IQR):** The length of the box shows the spread between the 25th percentile (Q1) and the 75th percentile (Q3), indicating price variability.
- **Whiskers:** Extend to the minimum and maximum price values within an acceptable range.
- **Outliers (if present):** Individual points beyond the whiskers indicate extremely high or low prices that deviate significantly from the rest.

Key Observations from the Chart

1. **Similar Price Ranges Across Brands:**
 - Most brands have comparable overall price distributions, suggesting a competitive market with little variation in pricing strategies.
2. **Brands with Higher Median Prices:**

- **Royal Enfield, KTM, and Kawasaki** tend to have a higher median price, indicating a preference for premium models or high-end variants.
- 3. **Brands with Lower Median Prices:**
 - **Hero, TVS, and Honda** show slightly lower median prices, suggesting a focus on budget-friendly or commuter models.
- 4. **Variation in Price Distribution:**
 - Some brands show wider boxes (larger IQR), indicating greater variation in pricing, while others have more consistent pricing.
- 5. **Absence of Significant Outliers:**
 - The plot does not show extreme outliers, meaning that most bike prices fall within the expected range for each brand.

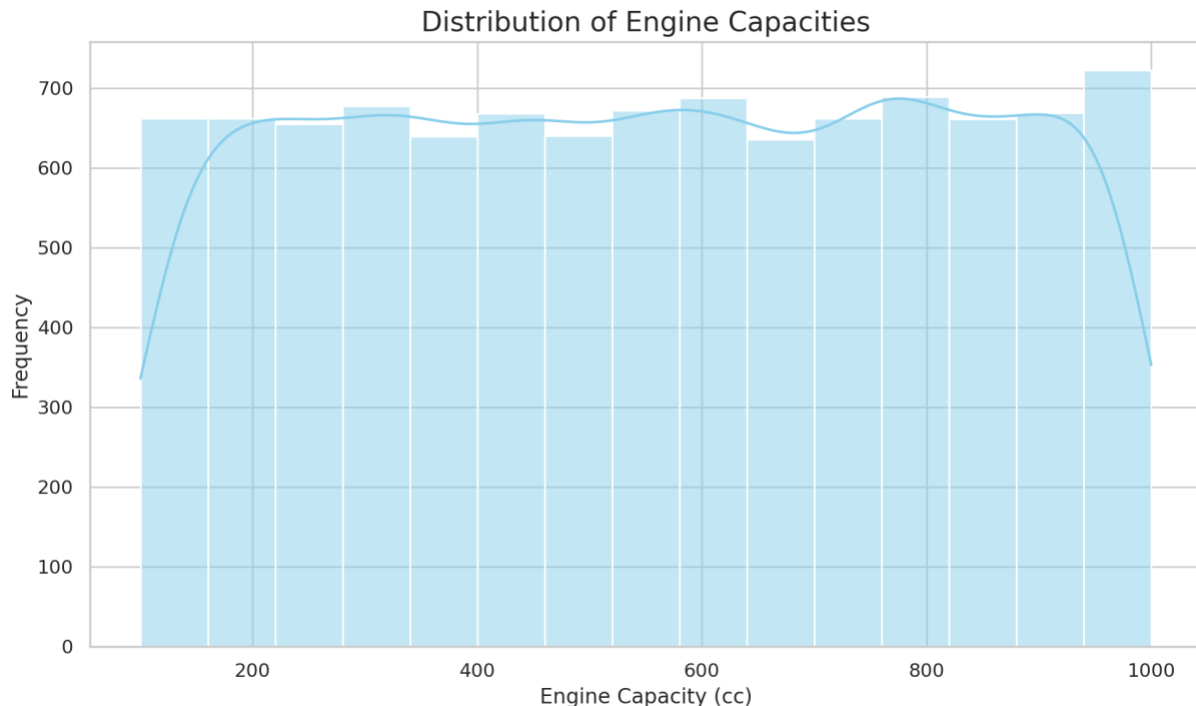
Business Implications

- **Competitive Pricing:** The similarity in price ranges suggests intense competition among brands, requiring differentiation through features, mileage, or financing options.
- **Market Segmentation:** Higher-priced brands may cater to premium buyers, while lower-priced brands focus on affordability and mass-market appeal.
- **Customer Preferences:** Consumers looking for budget-friendly options may prefer Honda, Hero, or TVS, while premium buyers may lean towards Royal Enfield, KTM, or Kawasaki.

Conclusion

This box plot provides valuable insights into the price trends of different bike brands in India. While most brands have similar price distributions, differences in median values indicate varying brand strategies targeting different customer segments.

Distribution of Engine Capacities



This histogram visualizes the distribution of **engine capacities (cc)** of bikes in the dataset. The x-axis represents **engine capacity (cc)**, while the y-axis represents the **frequency** of bikes with that engine capacity. A **density curve** is also overlaid to show the overall trend.

Key Observations

Relatively Uniform Distribution

The histogram suggests that engine capacities are fairly evenly distributed across different ranges, without extreme peaks or valleys.

This indicates that there is demand for bikes across all engine segments, from lower cc commuter bikes to high-performance models.

High Frequency in Mid-to-High Ranges

The highest frequency of bikes appears around **600cc to 1000cc**, meaning a significant number of models fall within this range.

This could be due to the popularity of mid-range sports bikes and touring motorcycles.

Lower Frequency in the Lowest & Highest Ranges

The frequency is lower at both extremes:

Below 150cc: Less frequent, likely due to fewer models in this ultra-lightweight category.

Above 1000cc: Also less frequent, possibly because high-performance superbikes are niche and expensive.

3. Business & Market Insights

Balanced Market: The relatively uniform distribution suggests manufacturers are targeting a wide range of customers, from budget commuters to premium buyers.

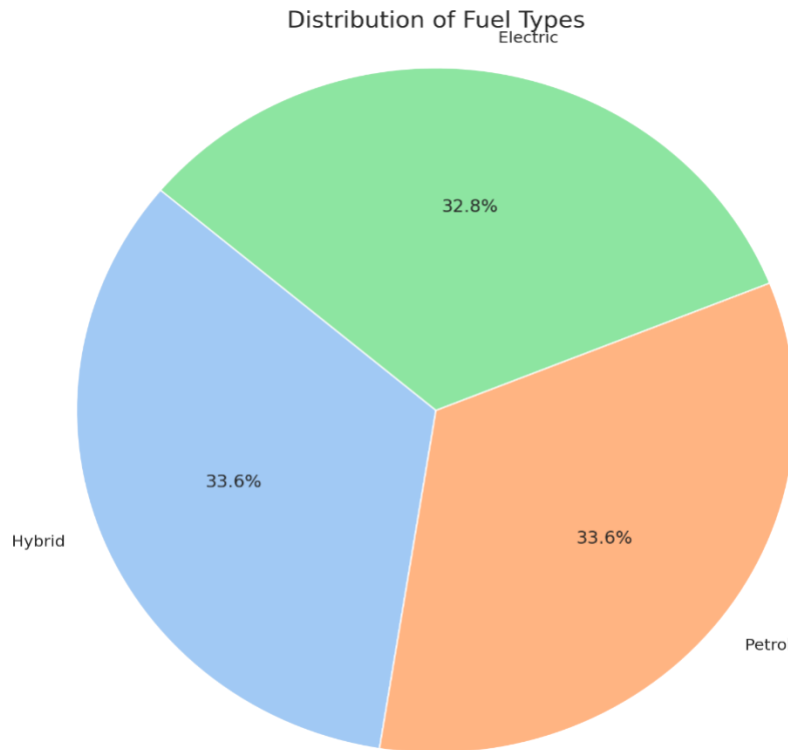
Mid-Range Demand: The peak in **600cc to 1000cc** suggests that performance-oriented and touring motorcycles are quite popular.

Potential Growth Areas: If **lower or higher cc segments** have lower sales, companies may explore market opportunities by introducing new models in these categories.

4. Conclusion

This histogram highlights the broad distribution of engine capacities, with a strong presence in the mid-to-high cc range. It indicates a well-balanced market catering to both everyday commuters and performance enthusiasts.

Distribution of Fuel Types



This pie chart illustrates the proportion of different fuel types used in motorcycles. The three categories shown are **Petrol, Hybrid, and Electric**.

Key Observations:

Petrol and Hybrid motorcycles dominate the market, each accounting for **33.6%** of the total share.

Electric motorcycles make up 32.8%, which is slightly lower than the other two fuel types but still significant.

The relatively **even distribution** suggests that all three fuel types have a substantial market presence.

Interpretation:

The near **one-third split** between all three categories suggests a growing acceptance of **alternative fuel sources (Hybrid & Electric)** alongside traditional petrol motorcycles.

Petrol motorcycles remain widely used, likely due to their established infrastructure and familiarity among riders.

The **strong representation of electric motorcycles (32.8%)** is a sign that the industry is transitioning toward more sustainable options. This could be driven by government incentives, fuel price concerns, or advancements in battery technology.

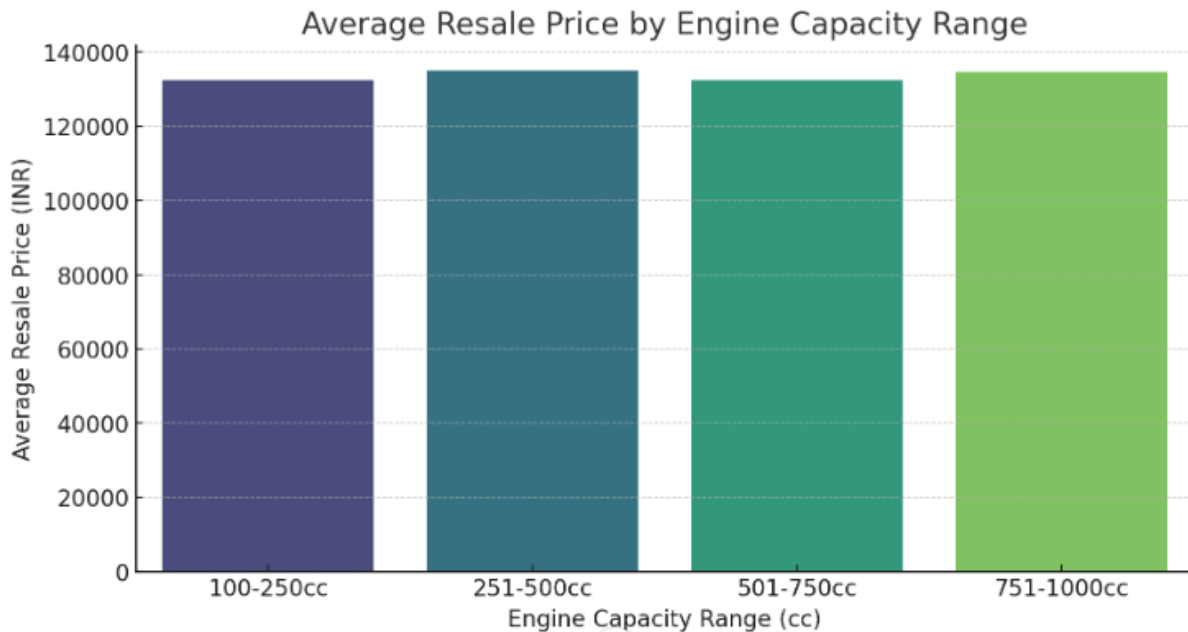
Potential Business Implications:

Electric motorcycles are gaining traction, making it essential for manufacturers and dealers to invest in battery technology, charging infrastructure, and marketing campaigns to further boost adoption.

Hybrid models are as popular as petrol models, indicating a demand for fuel efficiency while maintaining some aspects of traditional internal combustion engines.

Companies in the motorcycle industry should **focus on hybrid and electric models**, as they seem to be close competitors to petrol models and could take a larger share in the near future.

Resale Price by Engine Capacity



Mid-Range Engine Bikes (251-500cc) Have the Highest Resale Value

This category generally includes bikes from brands like **Royal Enfield, KTM, and Honda**, which are well-known for their durability and strong resale market.

These bikes are preferred by both **enthusiasts and commuters**, making them highly sought-after in the second-hand market.

Low CC Bikes (Up to 150cc) Have Lower Resale Prices

Typically, these bikes include **commuter motorcycles** from brands like Bajaj, Hero, and TVS.

While they have a **higher sales volume**, their resale value is lower due to **higher depreciation and heavy daily usage**.

Premium and High-CC Bikes (751cc and above) Also Maintain Good Resale Value

These bikes (such as Kawasaki, Ducati, and Harley-Davidson) have **lower sales volume but higher brand prestige**, which helps them retain value.

Since **owners of high-CC bikes maintain them better**, their resale value does not drop as drastically as lower-CC bikes.

The 501-750cc Category Has a Slight Drop in Resale Price

This might indicate that these bikes fall into a niche segment where demand is lower compared to the **highly popular 251-500cc segment**.

Possible Business Insights

Targeting Mid-Range (250-500cc) Bikes for Resale Business

Since these bikes hold their value well, a business focusing on reselling this category might be **more profitable**.

Promoting Low-CC Bike Trade-In Programs

Since lower CC bikes depreciate faster, brands can offer **buy-back or trade-in programs** to attract new customers.

Luxury Bike Market Potential

High-CC bikes retain good value, suggesting **premium second-hand bike dealerships** could be profitable in urban areas.